

FA0001_Sakana_v2.pdf COMPLETED

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Search

CONSENSUS SCORE

2/6

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ORIGINALITY

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REVIEWERS

Consensus

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GPT 5.4

2/6

Gemini 3.1 Pro

1/6

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Consensus review synthesized from 2 independent reviewers

Consensus
2/6

GPT 5.4
2/6

Gemini 3.1 Pro
1/6

Both reviewers recommend rejection and agree on the central diagnosis: the paper presents an interesting high-level idea but not a sufficiently specified or validated method. There is broad agreement that the submission lacks a formal algorithm, omits critical experimental details, provides no convincing quantitative evidence or baseline comparisons, and therefore does not support its claims about mitigating emergent misalignment while preserving task performance. Both also note internally corroborated presentation problems, including a missing Figure 1 and an implementation inconsistency between the Abstract (numpy/standard Python) and Section 4.1 (PyTorch). The main divergence is in how much credit to give the underlying concept: Reviewer 1 sees modest promise in the canary-controlled feedback idea and rates clarity/significance/originality slightly higher, whereas Reviewer 2 views the work as essentially incomplete and raises academic-integrity concerns. On adjudication, the concept itself merits limited novelty/clarity credit, but the demonstrated contribution remains too weak for acceptance. No external research context was provided, so claims about fabricated citations cannot be independently verified here and should be treated as concerns requiring separate checking rather than established fact.

STRENGTHS

The strongest aspect of the submission is the high-level proposal: using periodic canary evaluations as a feedback signal to adapt the ratio of safe data to task-specific data during fine-tuning is a plausible and potentially useful design pattern. Both reviewers agree that the paper targets an important problem—emergent misalignment from narrow fine-tuning—and that the main intuition is easy to grasp. The manuscript also appears conventionally organized and readable at a coarse level, enough for the intended contribution to be understood even though the technical realization is missing.

WEAKNESSES

The weaknesses are substantial and decisive. The method is under-specified: there is no mathematical formulation, pseudocode, update rule, thresholding scheme, evaluation cadence, initialization, clipping/smoothing policy, or precise definition of the canary metric. The empirical section is also inadequate: there are no convincing quantitative results, tables, ablations, or direct baseline comparisons despite claims that such baselines exist. The paper does not validate its key assumption that benign canary prompts are a reliable proxy for harmful emergent misalignment, which is central to the entire approach. Reproducibility is very poor because the synthetic dataset, prompt templates, model dimensions, hyperparameters, seeds, and other setup details are largely absent. Even within the reported to

evaluation setting itself is narrow—a synthetic dataset with an LSTM—so the broader claims about language-model alignment are unsupported. Finally, the paper contains internal inconsistencies and apparent omissions, notably the missing Figure 1 and the numpy/PyTorch contradiction. Reviewer 2 additionally alleges fabricated citations, but that claim cannot be independently confirmed from the provided context.

QUESTIONS FOR AUTHORS

Please provide explicit pseudocode or a mathematical formulation for the controller, including the initial safe/task ratio, evaluation interval, canary statistic, intervention threshold, update magnitude, and any clipping or smoothing.

What exactly are the task, safe, and canary prompts? Please include concrete examples, generation templates, dataset sizes, and train/validation splits.

Why should benign canary prompts diagnose emergent misalignment? Please add an experiment showing correlation between the canary metric and a separate held-out harmful-behavior or misalignment evaluation.

Where are the quantitative results and baseline comparisons promised in the text (for example, fixed-ratio interleaving or KL-style regularization)? Please report means and variance across multiple seeds.

Please define the reported metric mathematically and separate task-performance metrics from safety metrics.

Please report the full experimental configuration: model size, tokenizer/vocabulary, optimizer, learning rate, batch size, sequence length, training steps, seeds, and hardware.

Please include the missing Figure 1 and, ideally, a plot of the interleaving ratio over time to show that the controller actually adapts during training.

Please resolve the inconsistency between the Abstract's numpy/standard Python description and Section 4.1's PyTorch description.

If the bibliography contains placeholder or incorrect references, please correct them and ensure all citations are real and properly matched to the claims they support.

The paper's conclusions are sharply limited by both the missing methodological detail and the narrow evaluation setting. Because the experiments are reported only on a synthetic dataset with a simple LSTM, the work does not establish relevance to modern language-model fine-tuning. More importantly, the submission does not validate the central assumption that benign canary prompts track harmful emergent misalignment. Without baseline comparisons, ablations, variance estimates, and complete implementation details, the robustness and even the basic existence of the claimed effect cannot be assessed. Any claims about safety mitigation, preserved coherence, or scalability should therefore be treated as preliminary at best.

DIMENSION ANALYSIS

Quality

1/4

Consensus is strong that technical quality is poor. Both reviewers assign the lowest score and give compatible reasons: the core mechanism is not defined precisely enough to assess or reproduce, the key proxy assumption about canary prompts is unvalidated, and the experimental claims are unsupported by adequate evidence. The internally corroborated issues—missing figure, inconsistent implementation description, absent baselines, and missing setup details—further weaken confidence. Even setting aside the unverified citation concerns, the paper does not meet the standard for a technically sound submission.

Clarity

2/4

The reviewers differ slightly (2 vs. 1). The paper seems readable enough for a reader to understand the broad idea, and its organization is conventional, which argues against the harshest clarity score. However, the submission omits so many definitions, metrics, and implementation details that expert scrutiny and reproduction are impossible. Adjudicating between the two, the paper is best described as understandable at a high level but technically unclear in the places that matter most.

Significance

1/4

The reviewers differ modestly (2 vs. 1). The problem being addressed is important, and the idea of probe-driven adaptive data mixing could be useful if properly developed. However, significance in the rubric depends on the demonstrated contribution, not just the topic area. Because the submission provides no convincing evidence that the method works, no realistic benchmark, and no validation of its central proxy, its current impact on the field is minimal. The concept has potential, but the paper as submitted does not yet advance understanding in a substantive way.

Originality

2/4

The reviewers again differ slightly (2 vs. 1). The most charitable reading is that the paper offers some conceptual novelty in framing canary prompts as an online control signal for adjusting safe/task data interleaving, rather than using a fixed mixing ratio. At the same time, the submission does not clearly differentiate itself from adjacent ideas such as adaptive data selection, curriculum scheduling, or monitoring-based intervention, and the related-work discussion appears weak. On balance, the idea deserves modest originality credit at the conceptual level, but not more than that.

ETHICAL CONCERNS

Yes. The clearest concern is overclaiming safety effectiveness from a minimally validated

misalignment. There are also integrity-related concerns raised by one reviewer regarding potentially fabricated citations; this cannot be independently verified from the provided context, but if confirmed it would be serious.